

8 Optimal Unemployment Insurance

This section is based on Chapter 21 from Ljungqvist and Sargent (2012), Shavell and Weiss (1979), and Hopenhayn and Nicolini (1997).

Earlier in the course we looked at the decisions of the agent regarding search intensity in a simple search model. We showed that an increase in the unemployment benefits turns into a reduction in the search intensity. Now we will extend this model by including an unemployment insurance agency designing an *optimal* unemployment insurance contract.

Environment

An unemployed worker has preferences over consumption and search effort $\{c_t, a_t\}_{t=0}^{\infty}$ given by

$$E \sum_{t=0}^{\infty} \beta^t [u(c_t) - a_t],$$

where $u(c)$ is strictly increasing, strictly concave and C^2 , and $c_t \geq 0, a_t \geq 0$. We assume that $u(0)$ is well defined (finite).

An unemployed worker searches with effort a and with probability $p(a)$ receives a permanent job at the beginning of the next period. Once a worker has found a job, he is beyond the grasp of the unemployed insurance agency. Assume that $a = 0$ for an employed worker. Assume for simplicity that all jobs are alike and offer the same wage $w \Rightarrow \frac{u(w)}{1-\beta}$ is the present discounted value.

We impose the following assumptions on $p(a)$:

- $p'(a) > 0, p''(a) < 0$
- $p(a) \in [0, 1]$
- $p(0) = 0$
- $\lim_{a \rightarrow 0} p'(a) = \infty$ and $\lim_{a \rightarrow \infty} p'(a) = 0$ to ensure an interior solution

The consumption good is non-storable. The unemployed worker has no savings and cannot borrow or lend.

Incentive Problem

a_t is not observable but the duration of the unemployment spell is observable. The insurance agency designs a contract: $c(\tau)$ = consumption of the unemployed worker as a function of the unemployment duration, that minimizes the agency's costs. The value for the employed worker is $V^e = \frac{u(w)}{1-\beta}$.

Autarky

In the absence of the insurance agency, the worker solves

$$V_{aut}^u = \max_{a \geq 0} \{u(0) - a + \beta[p(a)V^e + (1 - p(a))V_{aut}^u]\}$$

$$\text{F.O.C.: } \beta p'(a)[V^e - V_{aut}^u] = 1$$

Because there is no state variable, a is constant during the unemployment spell.

Unemployment Insurance with Full Information

Suppose the insurance agency can observe a .

Let V be a promised value to an unemployed worker, $C^*(V)$ the expected discounted costs of giving (at least) V to the worker, and $R = \beta^{-1}$. The insurance agency is risk neutral. We will look for the optimal unemployment insurance contract (solution to the planner's problem).

Bellman Equation

$$\begin{aligned} C^*(V) &= \min_{c, a, V^u} \{c + \beta[1 - p(a)]C^*(V^u)\} \\ \text{s.t.} \quad V &\leq u(c) - a + \beta\{p(a)V^e + [1 - p(a)]V^u\} \quad (\theta) \end{aligned}$$

$C^*(V)$ is strictly convex because a higher V implies a lower marginal utility of the worker. Hence, additional expected "utils" can be granted to the worker only at an increasing marginal cost in terms of the consumption good.

F.O.C.

$$[c] : \theta = \frac{1}{u'(c)} \quad (1)$$

$$[a] : C^*(V^u) = \theta \left[\frac{1}{\beta p'(a)} - (V^e - V^u) \right] \quad (2)$$

$$[V^u] : C^{*'}(V^u) = \theta \quad (3)$$

$$[Env] : C^{*'}(V) = \theta \quad (4)$$

From (3) and (4) plus the fact that $C^*(V)$ is strictly convex we conclude that $V^u = V$. Using this, (2) becomes

$$C^*(V) = C^{*'}(V) \left[\frac{1}{\beta p'(a)} - (V^e - V) \right],$$

and 1 becomes

$$C^{*'}(V) = \frac{1}{u'(c)}.$$

These equations give us c and a as functions of V . (Notice that as V rises, c rises and a falls.)

Since the value V is constant over the unemployment spell, both effort and consumption are also constant over the unemployment spell. Thus, the unemployed worker's consumption is fully smoothed. But the worker's consumption is not smoothed across states of employment and unemployment, unless $V = V^e$. (If $V > V^e$, or equivalently $u(c) - a > u(w)$, then the worker does not have incentives to accept the job offer.) If the agency could tax the worker during his employment this tax would be fixed.

Relative to autarky, the insurance agency gives higher c and asks for a lower a . But a is higher than the effort that the worker would choose himself if he was facing the full-information contract: From (1), (2), and the fact that $C^*(V^u) > 0$ (insurance is costly):

$$[\beta p'(a)]^{-1} > V^e - V^u.$$

If the worker were to choose a freely (we are considering a one-shot deviation here), he would choose a such that

$$[\beta p'(a)]^{-1} = V^e - V^u.$$

Thus, the worker has incentives to reduce the search effort relative to the socially-efficient one. So if the effort is not observable, this is a problem.

Unemployment Insurance with Asymmetric Information

Now a is unobservable, chosen accordingly to the worker's FOC: $\beta p'(a)[V^e - V^u] = 1$. The insurance agency has to design an optimal contract that respects this choice.

$$\begin{aligned} C(V) &= \min_{c,a,V^u} \{c + \beta[1 - p(a)]C(V^u)\} \\ \text{s.t.} \quad V &= u(c) - a + \beta\{p(a)V^e + [1 - p(a)]V^u\} \quad (\theta) \quad (PK) \\ \beta p'(a)[V^e - V^u] &= 1 \quad (\eta) \quad (IC) \end{aligned}$$

F.O.C.

$$[c] : \theta = \frac{1}{u'(c)} \quad (5)$$

$$[a] : C(V^u) = \theta \left[\frac{1}{\beta p'(a)} - (V^e - V^u) \right] - \eta \frac{p''(a)}{p'(a)} (V^e - V^u) \underbrace{=}_{\text{using (IC)}} - \eta \frac{p''(a)}{p'(a)} (V^e - V^u) \quad (6)$$

$$[V^u] : C'(V^u) = \theta - \eta \frac{p'(a)}{1 - p(a)} \quad (7)$$

$$[Env] : C'(V) = \theta \quad (8)$$

Given that $C(V^u) > 0$ and (6) we conclude that $\eta > 0$. Then using (7)-(8) we have that

$$C'(\underbrace{V^u}_{V_{t+1}}) = \frac{1}{u'(c_{t+1})} = \frac{1}{u'(c_t)} - \eta \frac{p'(a)}{1 - p(a)} \Rightarrow c_{t+1} < c_t$$

Also, (7)-(8) imply $C'(V^u) < C'(V) \Rightarrow V^u < V$ if C is convex (we can use loteries to convexify the value function). Hence, $V_{t+1} < V_t$.

(Another way to show that $c_{t+1} < c_t$:

$$\frac{1}{u'(c_t)} = C'(V_t)$$

and assuming that C is convex, and using concavity of u and the fact that $V_{t+1} < V_t$, we have that $c_{t+1} < c_t$. Notice however that unlike the first way, this one relies on the convexity of C .)

Then, from

$$\beta p'(a_t) \left[V^e - \underbrace{V^u}_{V_{i+1}} \right] = 1,$$

concavity of p and the fact that V_t is decreasing over time, it follows that a_t is increasing over time. The unemployment insurance agency prescribes a decreasing over time stream of consumption and continuation value to the unemployed worker, which induces him to search harder and harder as time passes.

References

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